

tf.compat.v1.debugging.assert_shapes

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https://www.tensorflow.org/api_docs/python/tf/compat/v1/debugging/assert_shapes

Assert tensor shapes and dimension size relationships between tensors.

Function Signatures

```
tf.compat.v1.debugging.assert_shapes( shapes, data=None,
summarize=None, message=None, name=None )
x = tf.zeros([n,q])
```

Code Examples

```
tf.compat.v1.debugging.assert_shapes(  
    shapes, data=None, summarize=None, message=None, name=None  
)
```

```
tf.compat.v1.debugging.assert_shapes(  
    shapes, data=None, summarize=None, message=None, name=None  
)
```

```
n = 10  
q = 3  
d = 7  
x = tf.zeros([n,q])  
y = tf.ones([n,d])  
param = tf.Variable([1.0, 2.0, 3.0])  
scalar = 1.0  
tf.debugging.assert_shapes([  
    (x, ('N', 'Q')),  
    (y, ('N', 'D')),  
    (param, ('Q',)),  
    (scalar, ()),  
])
```

```
x = tf.zeros([n,q])
```

```
y = tf.ones([n,d])
```

Documentation

Assert tensor shapes and dimension size relationships between tensors.

This Op checks that a collection of tensors shape relationships satisfies given constraints.

Example:

Example of adding a dependency to an operation:

If x, y, param or scalar does not have a shape that satisfies all specified constraints, message, as well as the first summarize entries of the first encountered violating tensor are printed, and `InvalidArgumentError` is raised.

Size entries in the specified shapes are checked against other entries by their hash, except:

- a size entry is interpreted as an explicit size if it can be parsed as an

integer primitive.

- a size entry is interpreted as any size if it is `None` or `'.'`.

If the first entry of a shape is `...` (type `Ellipsis`) or `'*'` that indicates a variable number of outer dimensions of unspecified size, i.e. the constraint applies to the inner-most dimensions only.

Scalar tensors and specified shapes of length zero (excluding the 'inner-most' prefix) are both treated as having a single dimension of size one.

- `('N', 'Q')` specifies a 2D shape wherein the first and second

dimensions of shape may or may not be equal.

- ('N', 'N', 'Q') specifies a 3D shape wherein the first and second

dimensions are equal.

- (1, 'N') specifies a 2D shape wherein the first dimension is

exactly 1 and the second dimension can be any value.

Note that the abstract dimension letters take effect across different tuple elements of the list. For example,

`tf.debugging.assert_shapes([(x, ('N', 'A')), (y, ('N', 'B'))])` asserts

that both `x` and `y` are rank-2 tensors and their first dimensions are equal (`N`).

shape can also be a `tf.TensorShape`.

Returns

Raises